

GFL Presence Agent Deployment Guide

Step-by-step instructions for IT staff to monitor which Windows computers are online across your organization using the FormGFL inventory system.

Document version: 1.0

Agent version: 1.0.0

Audience: IT administrators, system deployers

1. Overview

The **GFL Presence Agent** is a small Windows program that runs as a background service on each PC. It sends a short heartbeat to your FormGFL server so IT can see which machines are turned on, who is logged in, and whether they match inventory records.

What the agent does

Action	Detail
Starts on boot	Runs automatically as a Windows Service
Sends heartbeats	Immediately on start, then every 5 minutes
Identifies the PC	Stable machine ID, hostname, BIOS serial number
Links to inventory	Serial match → asset, employee, and branch in FormGFL
Privacy	Sends hostname, serial, OS, user, LAN IP only — no files or process data

Online / offline logic

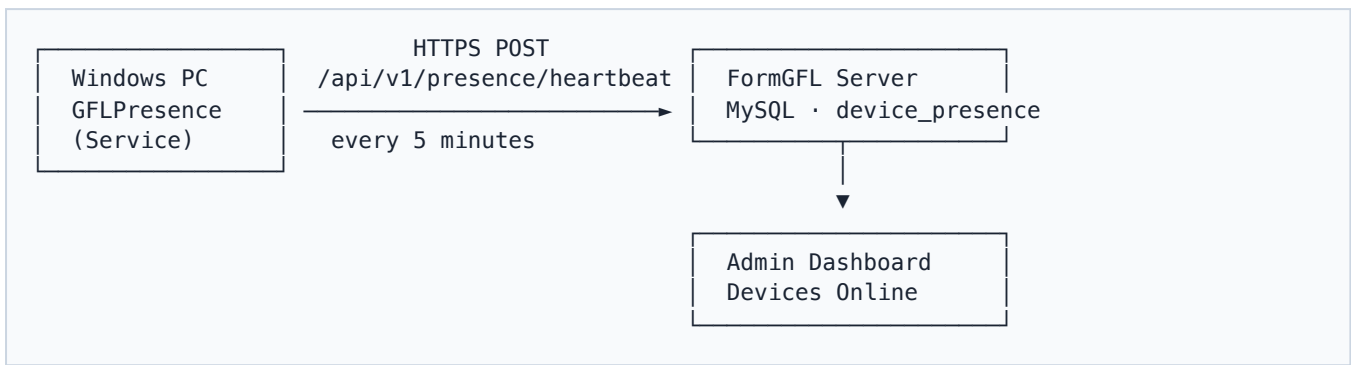
PC boots → Agent starts → POST heartbeat → FormGFL marks device **ONLINE**
PC off / no network → No heartbeat for 15 min → FormGFL marks device **OFFLINE**

View all devices: FormGFL Admin → System → Devices Online

2. Requirements

Location	Requirement
Each Windows PC	Windows 10+, outbound HTTPS, admin rights for install
FormGFL server	Deployed with presence monitoring; API key with <code>presence.report</code> + <code>presence.enroll</code>

3. Architecture



4. Phase 1 — Server Setup (FormGFL Admin)

Complete this **once** before installing agents on PCs.

Option A — Command line (recommended)

From the FormGFL project folder:

```
npm run seed:presence-key
```

Important: Copy the API key when printed — it is shown only once. Key name: **GFL Presence Agent** ·
Scopes: `presence.report` + `presence.enroll` · Whitelist: `0.0.0.0/0`

Option B — Admin web interface

1. Sign in to FormGFL admin
2. Go to **Developer** → **API Integrations**
3. Create a new API key with scopes **presence.report** and **presence.enroll**
4. Set IP whitelist to **0.0.0.0/0**
5. Copy the key (`gfl_...`)

Verify

Open FormGFL admin → **Devices Online**. Page should load (empty until agents are installed).

5. Phase 2 — Build the Agent

Build the `.exe` once on a developer machine, then copy to all PCs.

Step 1 — Install Go (if needed)

Download from <https://go.dev/dl/> or run `brew install go` on macOS.

Step 2 — Build

From the FormGFL project root:

```
npm run agent:build
```

Output: `agent/GFLPresence.exe` (approx. 6 MB)

6. Phase 3 — Prepare Configuration

Create `config.json` for deployment to each PC.

Location on PC: C:\ProgramData\GFLPresence\config.json

```
{
  "apiUrl": "https://your-formgfl-domain.com/api/v1/presence/heartbeat",
  "apiKey": "gfl_your_api_key_here",
  "intervalSeconds": 300
}
```

Field	Description
apiUrl	Full heartbeat URL — use HTTPS in production
apiKey	API key from Phase 1
intervalSeconds	Seconds between heartbeats (default: 300 = 5 min)

7. Phase 4 — Install with CMD Wizard (recommended)

This is the easiest path for end users. The wizard enrolls the PC, creates/links the inventory asset, and starts the service.

Step 4.1 — Prepare the install folder

Copy these files to a USB stick or network share:

```
GFLPresence/  
  Install-GFLPresence.cmd  
  setup.ps1  
  GFLPresence.exe  
  config.json
```

Pre-fill `config.json` with your production `apiUrl` and API key.

Step 4.2 — Run as Administrator

1. Right-click **Install-GFLPresence.cmd** → **Run as administrator**
2. Approve the UAC prompt

Step 4.3 — Answer the prompts

Prompt	What happens
Select branch	Lists active branches from FormGFL
Employee email	Looks up employee for that branch (must already exist)
BIOS serial	Auto-detected; confirm or type manually
Device type	Laptop / Desktop / All-in-One — used only when creating a new asset

Step 4.4 — What the installer does

- Creates the inventory asset if the serial is new, or reassigns an existing asset
- Marks the PC online in **Devices Online**
- Writes `C:\ProgramData\GFLPresence\config.json`
- Installs and starts the **GFL PC Presence Agent** Windows Service

Step 4.5 — Verify

Where	What to check
FormGFL → Devices Online	PC appears as Online with employee + branch
Inventory	Asset assigned to the employee
Services (<code>services.msc</code>)	GFL PC Presence Agent — Running, Automatic

7b. Manual install (advanced)

Use only if you do not need the enroll/link wizard.

```
New-Item -ItemType Directory -Force -Path "C:\Program Files\GFLPresence"
Copy-Item GFLPresence.exe "C:\Program Files\GFLPresence\"
New-Item -ItemType Directory -Force -Path "C:\ProgramData\GFLPresence"
Copy-Item config.json "C:\ProgramData\GFLPresence\config.json"
cd "C:\Program Files\GFLPresence"
.\GFLPresence.exe -service install
.\GFLPresence.exe -service start
```

8. Service Management Commands

Run from elevated PowerShell in `C:\Program Files\GFLPresence\`:

Command	Description
<code>GFLPresence.exe -service install</code>	Register auto-start Windows Service
<code>GFLPresence.exe -service start</code>	Start the service
<code>GFLPresence.exe -service stop</code>	Stop the service
<code>GFLPresence.exe -service restart</code>	Restart the service
<code>GFLPresence.exe -service uninstall</code>	Remove the service
<code>GFLPresence.exe run</code>	Foreground mode (testing only)

9. Linking PCs to Inventory

When the BIOS serial matches a product **SKU** in FormGFL, the device links to the asset, employee, and branch automatically.

Check BIOS serial on a PC:

```
(Get-CimInstance Win32_BIOS).SerialNumber
```

10. Rollout to Many PCs

Method	Notes
Group Policy (GPO)	Deploy exe, config, and one-time install script
Microsoft Intune	Package as Win32 app
SCCM / MECM	Standard software deployment

Item	Recommended path
Program files	C:\Program Files\GFLPresence\
Configuration	C:\ProgramData\GFLPresence\config.json
Machine ID (auto)	C:\ProgramData\GFLPresence\machine-id.txt

11. Security Notes

- Store API key only in `config.json` on each PC
- Use a dedicated key scoped to `presence.report` + `presence.enroll`
- Use HTTPS in production
- Agent sends hostname, serial, OS, username, LAN IP only
- Rotate the API key if a machine is decommissioned

12. Troubleshooting

Problem	Solution
Service won't start	Run <code>GFLPresence.exe run</code> ; check config JSON
401 Unauthorized	Verify API key has <code>presence.report</code> and <code>presence.enroll</code>
403 IP not whitelisted	Set whitelist to <code>0.0.0.0/0</code> in Admin → Developer
Employee not found	Email must exist for the selected branch
Not linked to employee	Re-run <code>Install-GFLPresence.cmd</code> as Administrator
Shows offline	Check HTTPS outbound; offline after 15 min without heartbeat
PC not appearing	Confirm service running; check Event Viewer

13. Deployment Checklist

- ☐ FormGFL server live; Devices Online page loads
 - ☐ API key created with `presence.report` + `presence.enroll` scopes
 - ☐ Install folder prepared (cmd, ps1, exe, config.json)
 - ☐ `Install-GFLPresence.cmd` run as Administrator
 - ☐ Branch selected and employee email verified
 - ☐ BIOS serial confirmed; asset created or linked
 - ☐ PC appears in Devices Online within 1 minute
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